

# CapSense: broadcast GPS position every 60 s

App-only setup. Starting point: node already on the **LONG\_MODERATE** preset, role **CLIENT**.

## 1 Find the right settings screen

The **Module configuration** list (MQTT, Serial, Telemetry, Neighbor Info...) is the **wrong half** — position settings are not there. Tap ← to go back one level.

Left rail > **Settings** > your node > the config list — two groups:

| Radio configuration                                                                                                                  | ← Position is in here            |
|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| User                                                                                                                                 |                                  |
| Channels                                                                                                                             | step 3                           |
| Device                                                                                                                               |                                  |
| <b>Position</b>                                                                                                                      | step 2                           |
| Power                                                                                                                                | step 4                           |
| Network · Display · LoRa · Bluetooth · Security                                                                                      |                                  |
| Module configuration                                                                                                                 | the screen you were on — skip it |
| MQTT · Serial · External Notification · Store & Forward · Range Test · Telemetry · Canned Message · Neighbor Info · Detection Sensor |                                  |

"Radio configuration" is the group heading above the list, not a row you tap — that is why you could not find it as a menu option. If your build does show it as a row, open it; Position is inside. Alternative entry point:

**Nodes** tab > tap your node > its settings.

## 2 Position — the heartbeat

| SETTING                       | VALUE              | WHY                                                                                                                                                                                                        |
|-------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GPS Mode</b>               | <b>Enabled</b>     | Not <i>Disabled</i> , not <i>Not present</i> .                                                                                                                                                             |
| <b>Position Broadcast (s)</b> | <b>60</b>          | The heartbeat. Default is 15 min. <input type="text" value="0"/> means "default", not "zero".                                                                                                              |
| <b>Smart Position</b>         | <b>OFF</b>         | ON by default. It adds <b>extra</b> transmissions whenever you move $\geq 100$ m (min 30 s apart), on top of the 60 s beacon. Off = exactly one packet per minute, which is what you budgeted airtime for. |
| <b>Fixed Position</b>         | <b>OFF</b>         | Would ignore the real GNSS.                                                                                                                                                                                |
| <b>GPS Update Interval</b>    | <b>30</b>          | Default is 2 minutes — leave it and every other beacon repeats stale coordinates. Must be $\leq 60$ .                                                                                                      |
| <b>Position flags</b>         | <b>Minimum</b>     | Each flag adds bytes to 1,440 packets/day.                                                                                                                                                                 |
| <b>GPS RX / TX pin</b>        | <b>Don't touch</b> | Set by the stock firmware.                                                                                                                                                                                 |

Tap **Save**. The node reboots and BLE drops for a few seconds — normal. Nothing applies until you save.

### 3 Channel

Radio configuration > **Channels** > primary channel (index 0)

| SETTING                               | VALUE   | WHY                                                                                                                                                                                                                                           |
|---------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Position precision / Location sharing | Precise | This is a 0-32 precision value, not a simple toggle: <b>0 = location never sent on this channel</b> , 32 = full precision, values in between snap you to a privacy grid. If it reads <i>Do not share</i> , the beacon carries no coordinates. |

Save. Do **not** change the channel name or PSK — the node would drop off your mesh.

### 4 Power

Radio configuration > **Power**

| SETTING      | VALUE | WHY                                                                         |
|--------------|-------|-----------------------------------------------------------------------------|
| Power Saving | OFF   | Sleep cycles cut the GNSS between beacons and make the heartbeat irregular. |

Save. That is the last configuration screen.

### 5 Verify

1. Outdoors, wait for the first fix — **1 to 5 minutes** on a cold start. The very first beacon after boot is skipped until a fresh fix exists.
2. **Nodes** tab > your CapSense: coordinates and a recent timestamp.
3. Watch **last heard** for 3-4 minutes: it must reset **every ~60 s**.
4. **Real check:** connect the app to a **different node** and confirm the CapSense shows up there refreshing every minute. That proves the LoRa path, not just Bluetooth.

### Summary

| SCREEN       | SETTING                    | VALUE                              |
|--------------|----------------------------|------------------------------------|
| Position     | GPS Mode                   | Enabled                            |
| Position     | Position Broadcast (s)     | 60                                 |
| Position     | Smart Position             | OFF                                |
| Position     | Fixed Position             | OFF                                |
| Position     | GPS Update Interval (s)    | 30                                 |
| Channels > 0 | Position precision         | Precise (32)                       |
| Power        | Power Saving               | OFF                                |
| LoRa         | Region · Modem Preset · TX | Yours · Long Range - Moderate · ON |
| Device       | Role                       | Client                             |

### Why it may still not beacon every 60 s

Two firmware rules override your setting. Neither shows an error in the app — the beacon just doesn't go out.

| RULE                     | EFFECT                                                                                                                                                                                                                                                                                                            |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Channel utilisation gate | Position packets are dropped while channel utilisation is $\geq 25\%$ for the Client role (Tracker roles get 40 %). A busy mesh silently stops your heartbeat.                                                                                                                                                    |
| Node-count scaling       | Above <b>40 online nodes</b> the firmware stretches the interval automatically. On LONG_MODERATE the factor is $1 + 0.16$ per extra node: <b>50 nodes</b> $\rightarrow \sim 2.6\times$ ( $\approx 158$ s), <b>60 nodes</b> $\rightarrow \sim 4.3\times$ ( $\approx 257$ s). You cannot hold 60 s on a large mesh. |

| SYMPTOM                    | FIX                                                                                           |
|----------------------------|-----------------------------------------------------------------------------------------------|
| Node visible, no position  | No GPS fix. Go outdoors, wait 5 min. Confirm GPS Mode = Enabled.                              |
| Beacons every 15 min       | That's the default — Save didn't apply. Reconnect, re-enter 60, Save.                         |
| Extra beacons while moving | Smart Position still ON.                                                                      |
| Same coordinates repeated  | GPS Update Interval > 60 s. Set it to 30.                                                     |
| Other nodes don't see it   | Channel position precision at 0, or mismatched region / preset / channel PSK, or TX disabled. |

#### COST ON LONG\_MODERATE

0.34 kbps: each position packet holds the channel for roughly 1.5–2 s, so one node alone sits near 2–3 % utilisation continuously, multiplied by every rebroadcast and hop — which feeds straight into the 25 % gate above. Battery with the GNSS always on lasts hours, not days. Use 60 s only on the node you are actively tracking; 120–300 s otherwise.

#### REVERT

Position Broadcast back to 900 s, Smart Position back ON. Field names follow the current Meshtastic app and firmware (Sept 2026); airtime figures are estimates for a typical position packet.